2 Answers to Team Problems

Answer 1. $-\frac{31}{6}$

Answer 2. $\sqrt{3}$

Answer 3. $(5, -1)$ and $(\sqrt{23}, 1)$

Answer 4. $\frac{51}{100}$

Answer 5. 37

Answer 6.

4	6	6	6
4	4	4	6
4	3	5	6
9	3	3	2

Answer 7. $\frac{20}{27}$

Answer 8. $\frac{119\sqrt{3}}{6}$

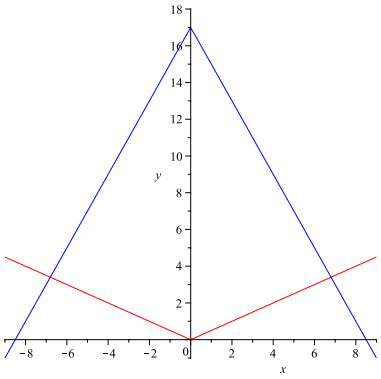
Answer 9. 240

Answer 10. 968

3 Solutions to Team Problems

Problem 1. Compute all values of a for which the intersection of the graphs of $y \geq \left|\frac{x}{2}\right|$ and $y \leq a|x| + 17$ is a region having area 51.

Solution 1. For $a < 0$, the graphs of the two functions are shown in the figure below.



In the first quadrant, the two bounding functions are $y = \frac{x}{2}$ and $y = ax+17$, so they intersect where $\frac{x}{2} = ax+17$. Moreover, because the triangle in the first quadrant is half of the total area it has area $\frac{51}{2}$. Taking the side along the y-axis as the base, the height is the x-value where the two lines intersect, hence $\frac{1}{2} \cdot 17 \cdot x = \frac{51}{2}$, thus $x = 3$. Now $\frac{3}{2} = 3a + 17$, so $a = -\mathbf{\frac{31}{6}}$.

Problem 2. Let $f(x) = \log_b x$ and let $g(x) = x^2 - 4x + 4$. Given that $f(g(x)) = g(f(x)) = 0$ has exactly one solution and that $b > 1$, compute b.

Solution 2. Consider the equation $f(g(x)) = 0$. Because $f(x) = \log_b(x)$, this equation holds precisely when $g(x) = 1$, no matter what b is. Setting $g(x) = x^2-4x+4$ equal to 1 yields the quadratic equation $x^2-4x+3 = 0$, which has two solutions, $x = 3$ and $x = 1$. Now consider the equation $g(f(x)) = 0$: $g(x)$ has one root at 2, so $f(x) = 2$, which implies that $x = b^2$. Therefore, in order for the equation $f(g(x)) = g(f(x)) = 0$ to have exactly one solution, b must equal $\mathbf{\sqrt{3}}$.

Problem 3. Compute all solutions to the following system of equations:

$$\begin{aligned} |y| - \frac{2x}{|x|} &= -1 \\ x|x| + y|y| &= 24. \end{aligned}$$

Solution 3. Note that x must be real because otherwise, the first equation would not be satisfied.

Case 1: x is positive. Then $\frac{2x}{|x|} = 2$, and the first equation yields $y = \pm 1$. The second equation then becomes $x^2 \pm 1 = 24$, so $x = 5$ (when $y = -1$) or $x = \sqrt{23}$ (when $y = 1$).

Case 2: x is negative. Then $\frac{2x}{|x|} = -2$, but in this case, the first equation becomes $|y| = -3$, which is impossible.

Thus the two solutions are $\mathbf{(5, -1)}$ and $\mathbf{(\sqrt{23}, 1)}$.

Problem 4. Two players are playing a game as follows. Integers N and M are each selected independently at random from the set $\{1, 2, \ldots, 50\}$. The players begin with an $N \times M$ grid of empty squares. A turn consists of filling in a single empty square, then filling in all the squares in the same row, and then filling in all the squares in the same column. The players alternate taking turns until all squares are filled in. Compute the probability that the player who goes first fills in the last square.

Solution 4. The outcome of the game is determined by the parity of $\min(N, M)$. This is because each time a player takes a turn, a row and a column that are previously not completely filled in (i.e., the row and column that contain the square selected on that turn) are now both completely filled in. Therefore, as soon as $\min(N, M)$ turns are taken, either all the rows will be completely filled, or all the columns, which means that the entire grid is filled in. Therefore the first player will fill in the last square if and only if $\min(N, M)$ is odd.

To compute the probability than $\min(N, M)$ is odd, count the number of pairs (N, M) satisfying this condition, and then divide by $50 \times 50 = 2500$, the total number of possible choices of N and M. First consider the case when $N < M$. There are 25 choices for N, from 1 to 49, and each results in $50 - N$ choices for M (i.e., the number of numbers larger than N). So there are $49 + 47 + \ldots + 3 + 1 = 25^2 = 625$ such pairs (N, M). The case when $M < N$ is symmetric, so there are another 625 cases there. Then there are another 25 cases where $N = M$, because N and M can equal any odd number between 1 and 50. This is a total of $625+625+25 = 1275$ pairs, so the desired probability is $\frac{1275}{2500} = \mathbf{\frac{51}{100}}$.

Problem 5. A sequence $T_1, T_2, \ldots$ is called (a, b)-*nacci* if $T_1 = a$, $T_2 = b$, and for $n \geq 3$, $T_n = T_{n-1} + T_{n-2}$. Compute the number of (a, b)-nacci sequences of positive integers such that $b > a$ and there exists an integer $k \geq 3$ with $T_k = 50$.

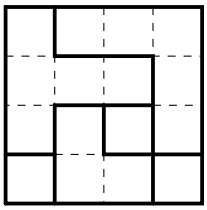
Solution 5. The sequence T_k must grow at least as fast as the $(1, 2)$-nacci sequence $1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$, thus for $k \geq 9$, $T_k > 50$.

If $k = 3$, then $a + b = 50$, which has 24 solutions with a being any integer from 1 to 24 inclusive.
If $k = 4$, then $a + 2b = 50$. Now a must be even and $a \leq 16$, so there are 8 solutions for $k = 4$.
If $k = 5$, then $2a + 3b = 50$. Now $a \equiv 1 \pmod 3$ and $a < 10$, so there are 3 solutions for $k = 5$.
If $k = 6$, then $3a + 5b = 50$. Now a must be a multiple of 5 and the only solution is $(5, 7)$.
If $k = 7$, then $5a + 8b = 50$. Now b must be a multiple of 5 and the only solution is $(2, 5)$.
If $k = 8$, then $8a + 13b = 50$, which has no solutions with $a < b$.

Thus the desired number of (a, b)-nacci sequences is $24 + 8 + 3 + 1 + 1 = \mathbf{37}$.

Problem 6. Complete the number puzzle below. Clues are given for the four rows. (Answers may not begin with a zero.) Cells inside a region must all contain the same digit, and each region contains a different digit.

1. Product of two primes
2. Multiple of 19
3. A perfect square
4. Multiple of 1-Across



Solution 6. Consider first 1-Across, which is of the form $\underline{A}\,\underline{B}\,\underline{B}\,\underline{B}$. 4-Across is a multiple of 1-Across, and of a different form, so they are not equal. Because both are four-digit numbers, A must be less than 5.

Next look at 2-Across. It is a four-digit multiple of 19 of the form $\underline{A}\,\underline{A}\,\underline{A}\,\underline{B}$, where A and B are the same digits

as above. Therefore 2-Across equals $1110A + B$, which is equivalent to $8A + B$ mod 19. Because 2-Across is a multiple of 19 and $A < 5$, the possibilities for (A, B) are $(2, 3)$ and $(4, 6)$.

Therefore 1-Across is either 2333 or 4666. The four-digit multiples of 2333 are 4666, 6999, and 9332. Only 9332 fits the pattern of 4-Across, so 4-Across is 9332. Because the digits in each region are distinct, 1-Across must be 4666.

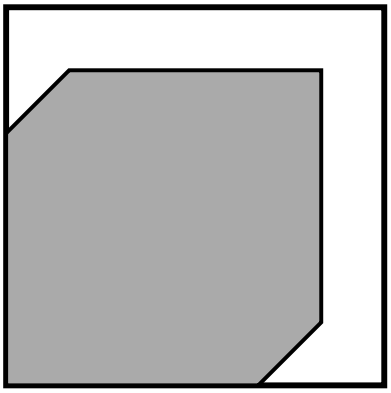
Then 2-Across is 4446. Filling in the digits that are the same in each region, 3-Across must be $43\underline{X}6$, where X is 0, 1, 5, 7, or 8, because these are the only unused digits. Noting that 3-Across is a perfect square and that $4356 = 66^2$, 3-Across is 4356.

4	6	6	6
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Problem 7. Real numbers x, y, and z are chosen at random from the unit interval $[0, 1]$. Compute the probability that $\max\{x, y, z\} - \min\{x, y, z\} \leq \frac{2}{3}$.

Solution 7. Consider points lying in a unit cube in the first octant with one corner at the origin. To find the volume of the region of points where the inequality holds, take cross-sections for each value of z from 0 to 1. For any particular value of z, the values of x and y must satisfy $0 \leq x \leq 1$, $0 \leq y \leq 1$, $-\frac{2}{3} \leq y - x \leq \frac{2}{3}$, $z - \frac{2}{3} \leq x \leq z + \frac{2}{3}$, and $z - \frac{2}{3} \leq y \leq z + \frac{2}{3}$.

Depending on the value of z, the resulting region is a square of area $\frac{4}{9}$ (when $z = 0$ or 1), a hexagon of area $\frac{4}{9} + \frac{4z}{3}$ (when $0 < z < \frac{1}{3}$), a hexagon of area $\frac{8}{9}$ (when $\frac{1}{3} \leq z \leq \frac{2}{3}$), or a hexagon of area $\frac{4}{9} + \frac{4(1-z)}{3}$ (when $\frac{2}{3} < z < 1$). The region where $z = \frac{1}{6}$ is shown.



In the cases where z is not between $\frac{1}{3}$ and $\frac{2}{3}$, the area changes linearly with z and hence can be averaged to $\frac{1}{2}\left(\frac{4}{9} + \frac{8}{9}\right) = \frac{2}{3}$. This area corresponds to $\frac{2}{3}$ of the z-values, with the other $\frac{1}{3}$ having area $\frac{8}{9}$, for a total probability of $\frac{2}{3}(\frac{2}{3}) + \frac{1}{3}(\frac{8}{9}) = \mathbf{\frac{20}{27}}$.

Problem 8. Square $ARML$ has sides of length 11. Collinear points X, Y, and Z are in the interior of $ARML$ such that $\triangle LXY$ and $\triangle RXZ$ are equilateral. Given that $YZ = 2$, compute $[RLX]$.

Solution 8. First note that X cannot be between Y and Z, because then the side lengths of triangles LXY and RXZ would each be less than $YZ = 2$, which would mean that XL and XR would each be less than 2, which is impossible because $XL + XR \geq LR = 11\sqrt{2}$. Without loss of generality, let Y lie between X and Z. Let $XY = YL = LX = x$, so that $XZ = ZR = RX = x + 2$. In triangle RLX, angle LXR has measure 120° because angles LXY and ZXR are each 60°. Applying the Law of Cosines then gives

$$\frac{x^2 + (x+2)^2 - (11\sqrt{2})^2}{2x(x+2)} = \cos 120^\circ = -\frac{1}{2}.$$

Solving this quadratic for x gives $x = -1 \pm \sqrt{\frac{241}{3}}$, and the positive solution for x is used because it is a side length. So $x = -1 + \sqrt{\frac{241}{3}}$ and $x + 2 = 1 + \sqrt{\frac{241}{3}}$, and the area $[RLX]$ is

$$\frac{1}{2}x(x+2)\sin 120^\circ = \frac{1}{2} \cdot \left(\frac{241}{3} - 1\right) \cdot \frac{\sqrt{3}}{2} = \mathbf{\frac{119\sqrt{3}}{6}}.$$

Problem 9. Compute the number of permutations $x_1, x_2, \ldots, x_{10}$ of the integers $-3, -2, -1, \ldots, 6$ that satisfy the chain of inequalities

$$x_1x_2 \leq x_2x_3 \leq \cdots \leq x_9x_{10}.$$

Solution 9. First note that no two negative numbers can occur consecutively because otherwise, either the resulting positive product would have to be followed by a non-positive product once the negatives are exhausted, or it would have to occur at the end but there are too many positive terms for that to be possible. Thus the sequence must start with alternating positive and negative values. Each inequality $x_ix_{i+1} \leq x_{i+1}x_{i+2}$ implies that $x_i < x_{i+2}$ if $x_{i+1} > 0$ and $x_i > x_{i+2}$ if $x_{i+1} < 0$ (the inequalities are strict because the x_is are distinct). The two main cases to consider are $x_1 = -3$ and $x_2 = -3$.

If $x_1 = -3$, then $x_3 = -2$, $x_5 = -1$, and either $x_6 = 0$ or $x_7 = 0$. If $x_6 = 0$, then the positive entries (x_2, x_4, x_7, x_8, x_9, and x_{10}) must satisfy $x_2 > x_4$, $x_7 < x_9$, and $x_8 < x_{10}$. There are $\frac{6!}{2!\cdot 2!\cdot 2!} = 90$ such sequences. If $x_7 = 0$, then the positive entries (x_2, x_4, x_6, x_7, x_8, and x_{10}) must satisfy $x_2 > x_4 > x_6$, and $x_8 < x_{10}$ for $\frac{6!}{3!\cdot 2!\cdot 1!} = 60$ more.

If $x_2 = -3$, then $x_4 = -2$, $x_6 = -1$, and either $x_7 = 0$ or $x_8 = 0$. If $x_7 = 0$, then the positive entries (x_1, x_3, x_4, x_8, x_9, and x_{10}) must satisfy $x_1 > x_3 > x_5$, and $x_8 < x_{10}$ for another 60. If $x_8 = 0$, then the positive entries (x_1, x_3, x_5, x_7, x_9, and x_{10}) must satisfy $x_1 > x_3 > x_5 > x_7$, for $\frac{6!}{4!\cdot 1!\cdot 1!} = 30$ more.

The desired number of permutations is therefore $90 + 60 + 60 + 30 = \mathbf{240}$.

Problem 10. Compute the greatest integer n for which the decimal representation of $\frac{1000}{n}$ is of the form $\underline{1}.\underline{T}\,\underline{A}\,\underline{S}\,\underline{T}\,\underline{Y}\,\underline{7}\ldots$, where the digits T, A, S, and Y are not necessarily distinct.

Solution 10. In order for the integral part of $\frac{1000}{n}$ to be 1, n must be between 501 and 1000. Let $m = 1000 - n$. Then

$$\frac{1000}{n} = \frac{1000}{1000 - m} = \frac{1}{1 - \frac{m}{1000}} = 1 + \frac{m}{1000} + \frac{m^2}{1000^2} + \frac{m^3}{1000^3} + \ldots.$$

For example, if $n = 998$, then $\frac{1000}{n}$ begins $1.002004008016032064\ldots$ and each block of 3 digits following the decimal point is a power of 2 until they begin to overlap.

To maximize n, first consider attempting to find a solution with $T = 0$, which implies $m < 100$, by the previous analysis, because the first block of 3 digits equals either m or m plus the thousands digit of m^2. Because the ones digit of the second block of 3 digits is 7, and m^2 cannot end in 7 (as no squares end in 7), the m^3 block must overlap into the m^2 block. Therefore $m^3 > 1000$, so $m > 10$.

Because $T = 0$ by (optimistic) assumption, and therefore the part of m^2 in the second block must have a zero in the hundreds place, the goal is to find four-digit squares with a zero in the hundreds place. (Note that m cannot have a two-digit square because $m > 10$, and the m^3 block will not overlap onto the 0 from the hundreds place of the second block as long as $m^3 < 100{,}000$.)

The first square with a zero in the hundreds place is $32^2 = 1024$, which corresponds to $m = 32$, $m^2 = 1024$, $m^3 = 32768$. Then the expansion for $\frac{1000}{1000-m}$ begins

$$\begin{aligned} & 1 + \frac{32}{1000} + \frac{1024}{1000^2} + \frac{32768}{1000^3} + \cdots \\ = \; & 1 + 0.032 + 0.001024 + 0.000032768 + \cdots \\ = \; & 1.033056768 + \cdots . \end{aligned}$$

This fits the pattern, except the sixth decimal digit is a 6 instead of a 7. However, $m^4 = 1024^2$ is just over 1 million, so its first digit overlaps onto the 6, making it a 7. Therefore $m = 32$, which corresponds to $n = \mathbf{968}$ giving the maximal n satisfying the given criteria. This can be confirmed by computing $\frac{1000}{968} = 1.033057\ldots$.

Note: The same method as in the solution can be used for negative integers m that are close to 0; the geometric series would have alternating signs. Also, replacing the 1000s with 10^ps yields a similar result, but instead of blocks of 3 digits apiece, the blocks will each have p digits.